



DEVICE RECOGNITION BASED ON BROWSER FINGERPRINTING

Şükrü Can Mayda
canmayda@marun.edu.tr

Nurbetül Çakır
nurbetulcakir@marun.edu.tr

Turgut Köroğlu
turgutkoroglu@marun.edu.tr

Supervisor: Assoc. Prof. Ömer Korçak



Introduction

Device fingerprinting is an advanced tracking technique that utilizes browser telemetry to create a unique digital identity without cookies. Browsers collect diverse system and hardware attributes to recognize instances through specific signals:

🔧 **Hardware Signals:** Screen resolution, CPU cores, GPU rendering (Canvas), and device memory profiles.

💻 **Software Signals:** User-Agent strings, installed fonts, browser extensions, and language settings.

🕒 **OS Signals:** Time zone offsets and operating system architecture.

Since these attributes fluctuate over time, linking tracking sessions to specific users requires a persistent mechanism. Our approach introduces a hybrid framework combining a **rule-based stage with a machine learning stage**. This integration leverages machine learning capabilities to enhance the performance and robustness of traditional rule-based fingerprint linking.

Dataset & Preprocessing

- 📄 228,773 fingerprint documents after preprocessing (281,155 raw)
- 📱 Each record is collected from desktop, mobile, smart TV devices
- 📋 96 attributes for linking: 86 categorical, 10 set-type (lists compared as sets)
- 🔒 IP address excluded (GDPR), 8 attributes dropped (low variety, high consistency)
- 🗑️ Empty/default dict, list, or string → normalized as missing
- 🔍 86,202 unique devices indicated by fingerPrintCookie
- ✂️ For preprocessing: duplicates removed, hierarchies flattened, repeated values inside set fields removed and sorted, field presence analyzed and bad records filtered or fixed

Shannon Entropy & Temporal Stability

Two measures define how much each attribute contributes to device discrimination:

📊 **Shannon Entropy:** Metric to measure how much information an attribute carries (**variety**).

$$H_{cat,raw}(A) = - \sum_{i=1}^j p(a_i) \log_2 p(a_i)$$

$$H_{set,raw}(S) = - \frac{1}{|\mathcal{E}_S|} \sum_{i=1}^{|\mathcal{E}_S|} [p_{s_i} \log_2 p_{s_i} + (1 - p_{s_i}) \log_2 (1 - p_{s_i})]$$

A : categorical attribute
 α : categorical attribute value
 S : set type attribute
 s : set type attribute value
 $\mathcal{E}_S$: set of all possible set type attribute values

🔄 **Temporal stability:** Metric to measure how often an attribute changes for the same device line (**consistency**).

$$J(S_{t-1}, S_t) = \begin{cases} 1 & S_{t-1} = S_t \\ \frac{|S_{t-1} \cap S_t|}{|S_{t-1} \cup S_t|} & S_{t-1} \neq S_t \end{cases}$$

J : Jaccard similarity
 τ : Change threshold

$$\Delta_{set,raw}(S; t-1, t) = \begin{cases} 1 & J(S_{t-1}, S_t) < \tau \\ 0 & J(S_{t-1}, S_t) \geq \tau \end{cases}$$

$$\Delta_{cat,raw}(A; t-1, t) = \begin{cases} 1 & v_t \neq v_{t-1} \\ 0 & v_t = v_{t-1} \end{cases}$$

$$Stab_{raw}(X) = 1 - \frac{\sum_{session} \sum_{t>1} \Delta_{raw}(X; t-1, t)}{N_{sessions}}, X \in \{A, S\}$$

Attributes that are both **high-entropy** and **high-stability** are the most useful for telling devices apart. Each attribute's weight combines its **normalized entropy and stability** with a parameter β .

Attribute & Type	Entropy (norm.)	Attribute & Type	Stability (norm.)
fonts (set)	1.00	math.atan (cat)	1.00
extensionParams. (set)	0.93	math.expm1 (cat)	1.00
architecture (cat)	0.73	vendorFlavors (set)	1.00
rendererUnmasked (cat)	0.71	priv.ClickMeasure. (cat)	0.99
canvas.text (cat)	0.55	math.cos (cat)	0.99

Table 1 Highest normalized entropy and stability attributes.

Rule Based Linking System

Each new fingerprint is matched against an active device list first using **strict**, then **relaxed** and finally **fallback blocking buckets** to limit candidates. Pairwise weighted attribute differences (categorical + set-valued) produce a similarity score; the best candidate below a threshold is **linked**, otherwise a **new device** is created. The flowchart summarizes this pipeline from incoming fingerprint to link / new device decision.

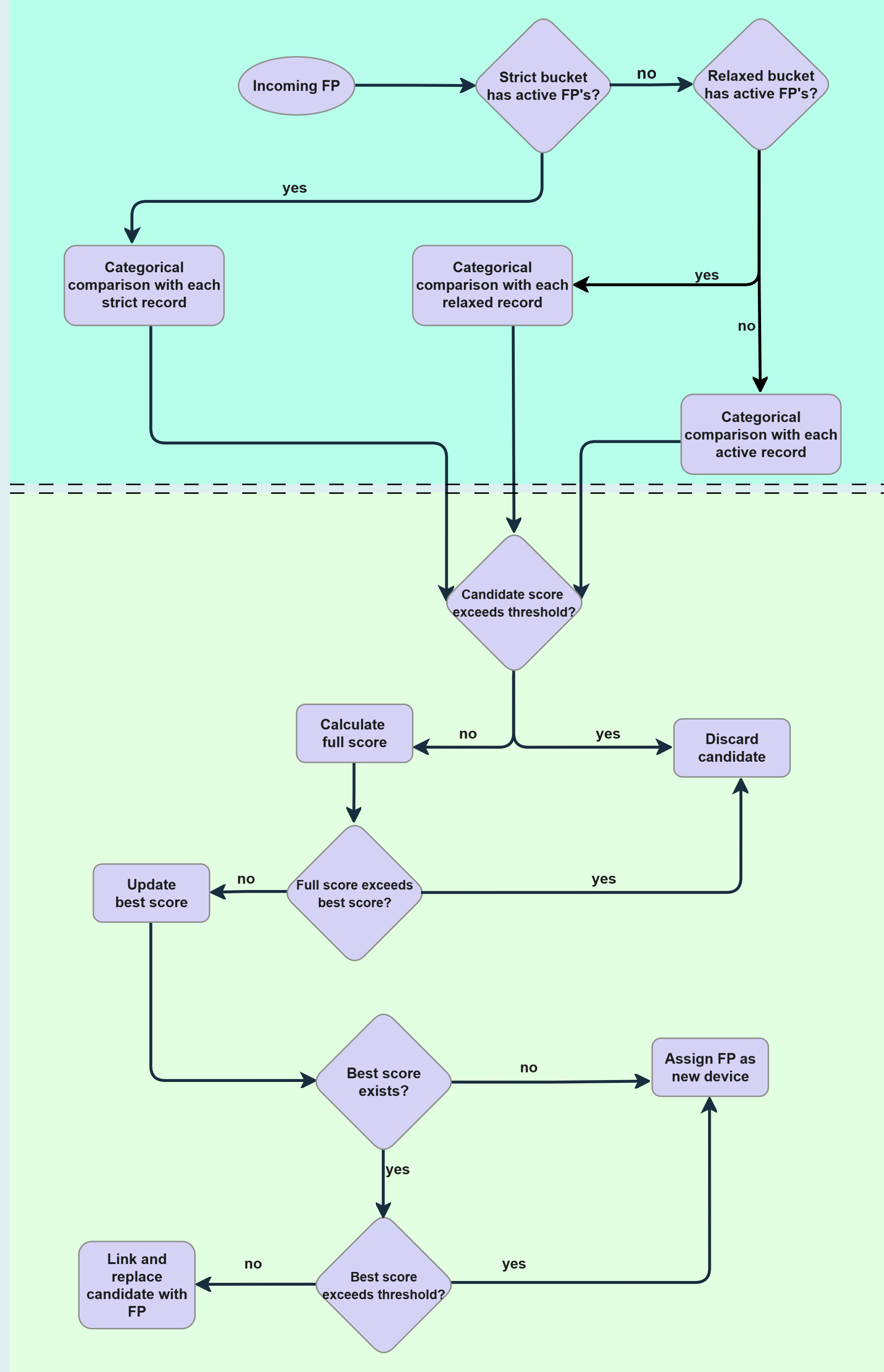


Figure 1 Rule-based candidate filtering, scoring, and link/new-device decision flow.

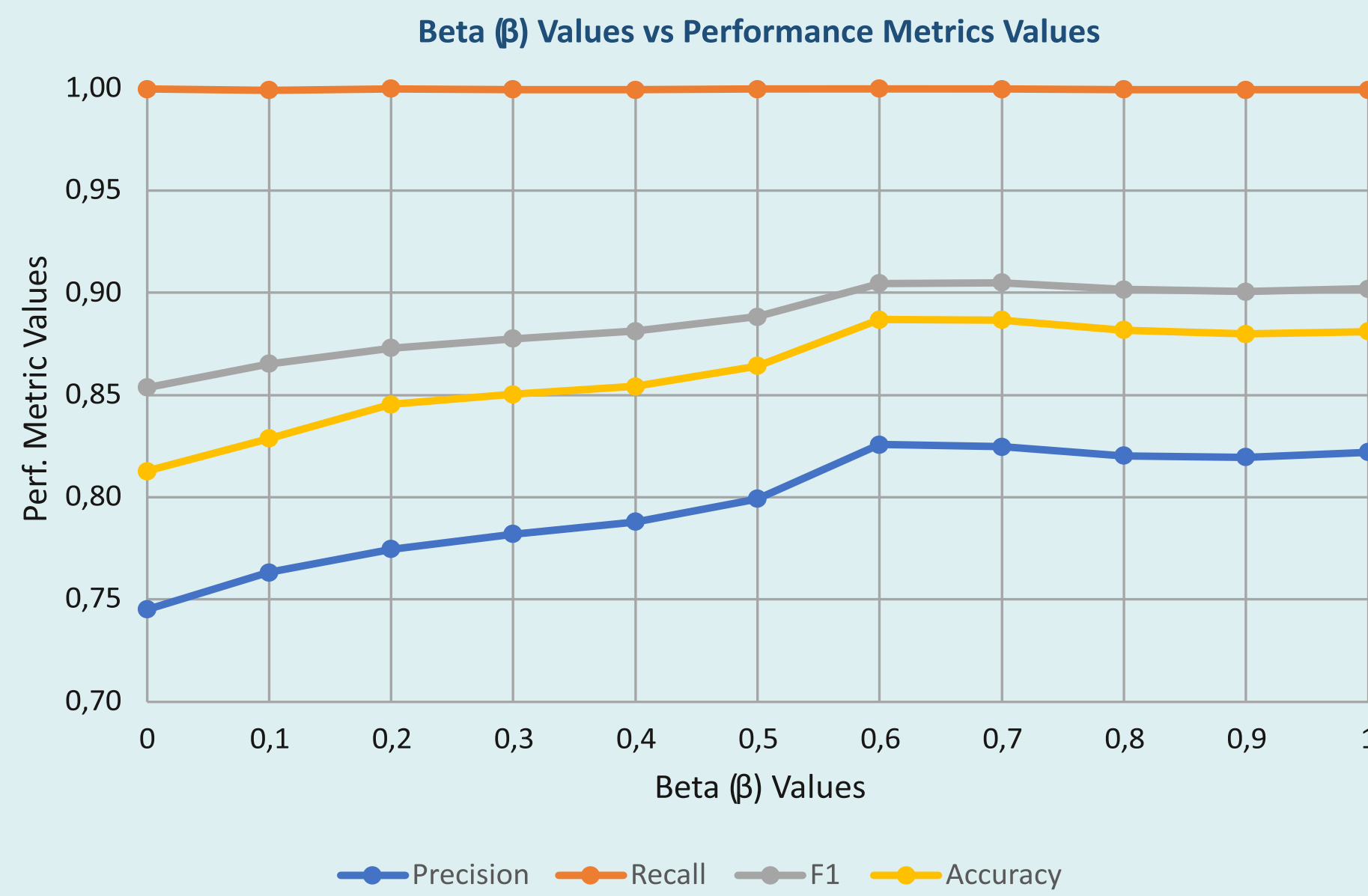


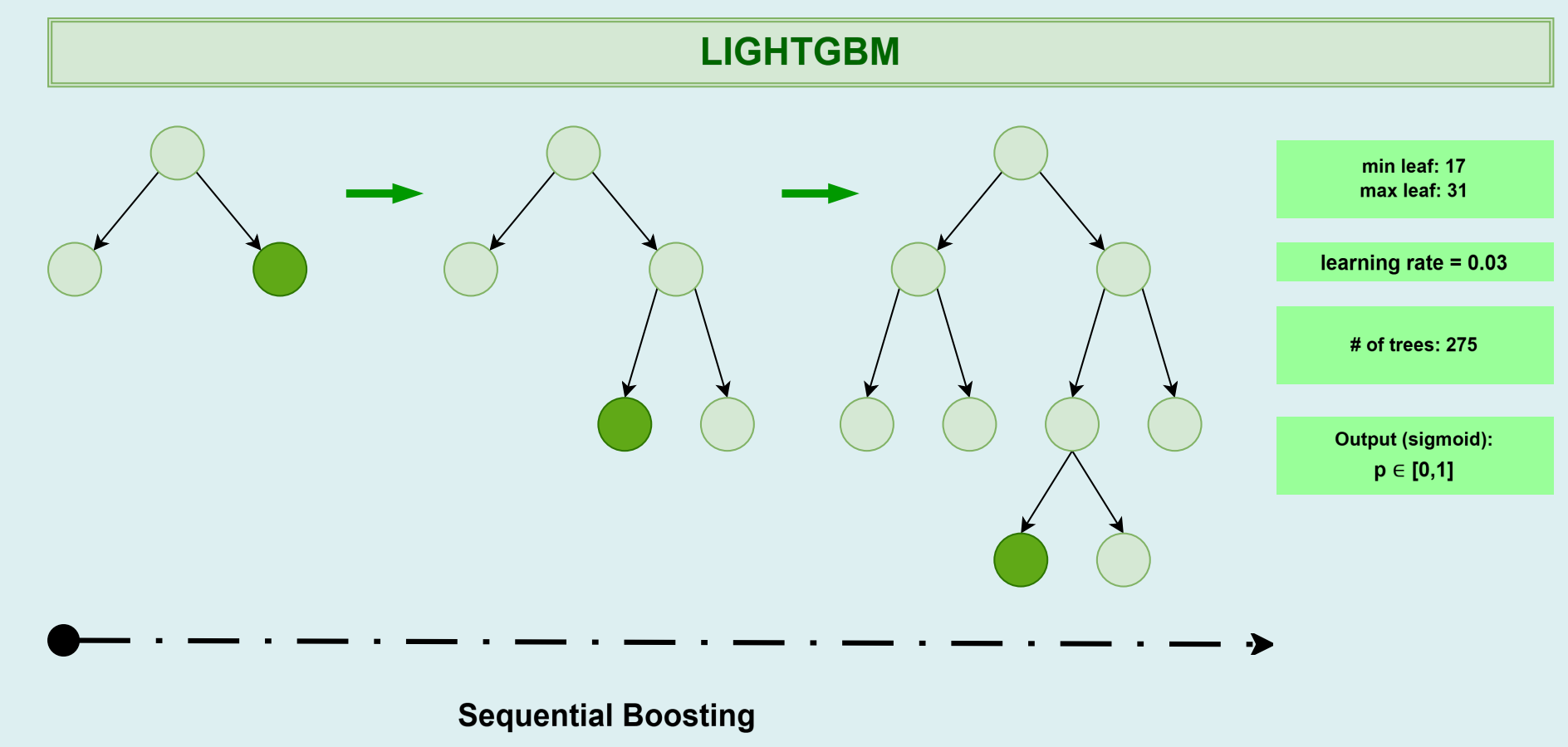
Figure 2 Rule-based metrics vs. beta (entropy-stability trade-off).

Machine Learning

The ML layer adds a **probability-based decision step** on top of the rule-based engine. Instead of using a fixed link rule, it evaluates comparison features generated from the **rule-based pairwise comparisons** and estimates whether a candidate fingerprint belongs to the same device. A **validation-tuned threshold** then converts this probability into a final linked or new device decision. The dataset contains **225,195 labeled samples** and **97 features**, created from the comparison outputs of the rule-based layer. These features include weighted attribute changes based on entropy and temporal stability and summary signals such as total difference score, number of changed features, threshold distance, time gap, highest weighted differences, and candidate rank. The dataset was split into **70% training**, **15% validation**, and **15% test sets**. Labels indicate whether two fingerprints share the same device. In this way, the model learns both detailed attribute-level changes and rule-based decision patterns to make more adaptive linking decisions.

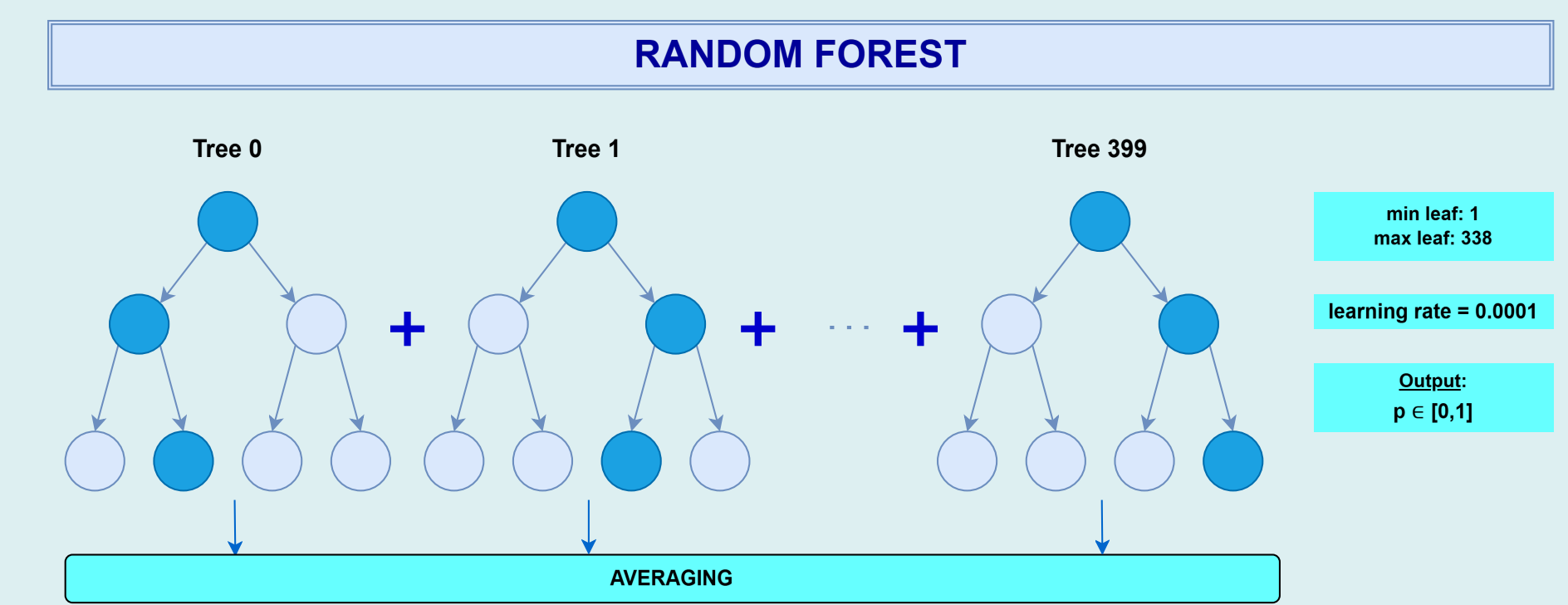
LightGBM

LightGBM learns which feature patterns indicate a true device match. For each incoming record, the model assigns link probabilities to the top candidates. The candidate with the highest probability is selected, and linking happens only if the confidence score passes the ML threshold.



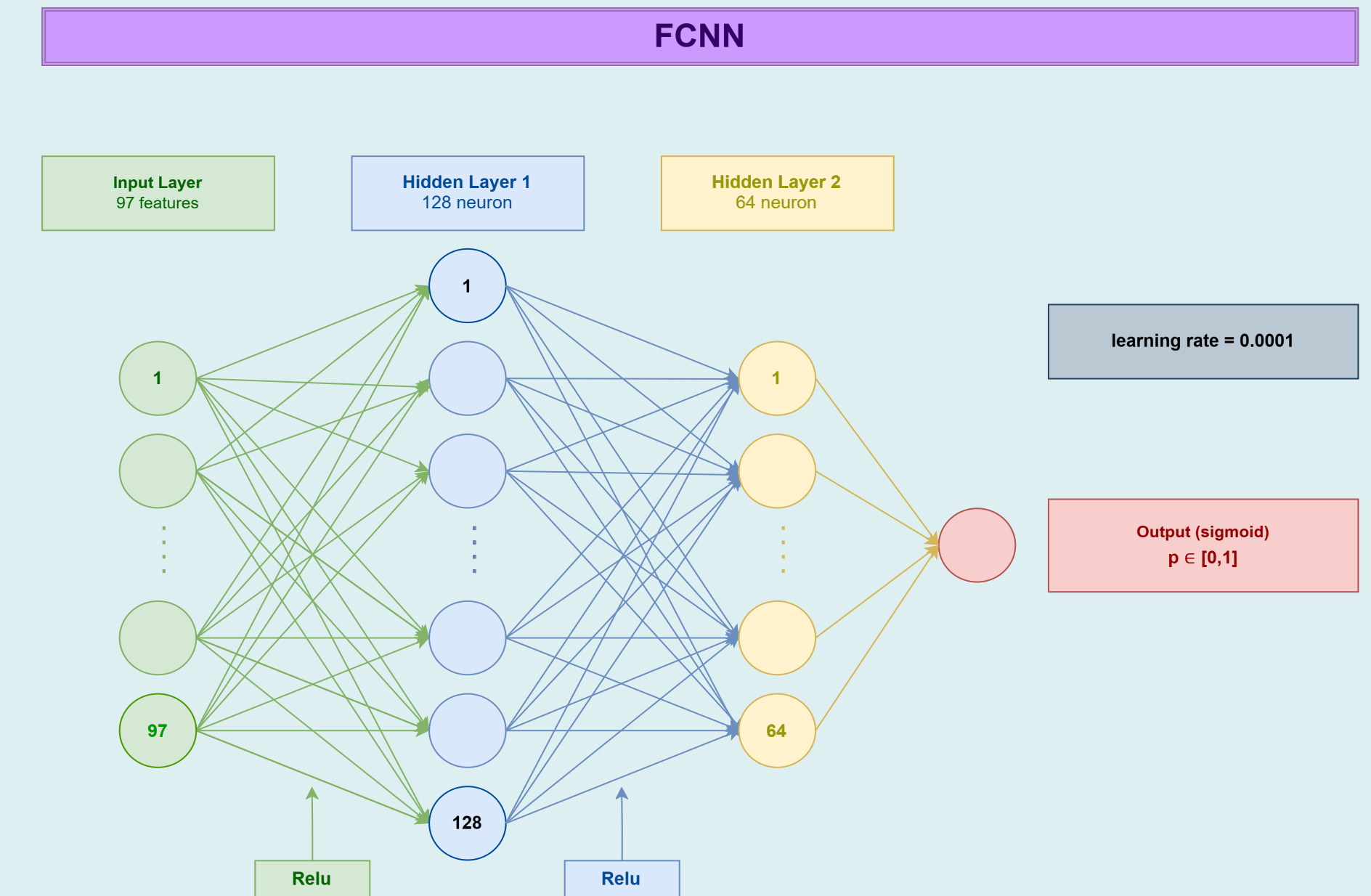
Random Forest

Random Forest uses random subsets of pair examples and features to train many decision trees. Each tree learns simple split rules on comparison features, and their votes are aggregated into a final link probability for each candidate. The candidate with the highest probability within the candidate group is selected as the best match.



FCNN

FCNN is a feedforward neural network with two hidden layers that learns non-linear match patterns. During inference, the network outputs a link probability for each candidate pair. The candidate with the highest probability is chosen and accepted only if it passes the decision threshold.



Conclusion

This project developed a two-stage hybrid framework for browser fingerprint linking. The rule-based layer provides fast and explainable candidate generation by comparing fingerprints through weighted attribute differences, while the machine learning layer re-evaluates these candidates using rule-based comparison features and probabilistic decision-making. The main contribution of the project is that it does not rely only on fixed rules or raw fingerprint data. Instead, it combines the **interpretability of rule-based logic** with the **adaptability of machine learning**. As a result, the system offers a **more balanced, reliable, and scalable approach** for linking browser fingerprints over time.

Future work may focus on improving threshold tuning, analyzing long-term tracking performance, testing the system on more diverse datasets, and extending the framework to broader browser and device environments.

References

- [1] P. Eckersley, "How Unique Is Your Web Browser?" in Proceedings of the Privacy Enhancing Technologies Symposium (PETS), Berlin, Germany, 2010, pp. 1–18.
- [2] A. Vastel, P. Laperdrix, and B. Baudry, "FP-Stalker: Tracking Browser Fingerprint Evolutions," in Proc. IEEE Symposium on Security and Privacy, San Francisco, CA, USA, 2018, pp. 1–17.
- [3] G. Pugliese, C. Cicconetti, and G. Stea, "Long-Term Observation on Browser Fingerprinting: Users' Trackability and Perspective," Proc. Privacy Enhancing Technologies (PoPETs), vol. 2020, no. 4, pp. 67–85, 2020.
- [4] A. A. Salomatin, "Web User Identification Based on Browser Fingerprints," Procedia Computer Science, vol. 190, pp. 125–132, 2021.
- [5] E. E. Şamlıoğlu, S. E. Taşçı, and B. Şamlıoğlu, "ThresholdFP: Enhanced Durability in Browser Fingerprinting," IEEE Access, vol. 13, pp. 1–17, 2025.

Used Technologies

